

LAB Kerberos NFS 설정 runbook

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This is the single current LAB runbook. It adapts the FARM NAS / AD / Kerberos model to LAB. LAB does not use Synology NAS for this path, so the Synology-specific join and DSM steps are replaced with ordinary Linux storage server AD member, RFC2307, and NFS server setup.

Superseded setup notes remain available in Git history. Reproduction procedures and captured outcomes remain under `../test/lab-kerberos-poc/`; they do not override the desired state in this file.

The goal is to apply the AD/Kerberos model first to the LAB `test_krb` export only. Existing production LAB paths should not be converted until this test path passes all identity, Kerberos, group, and container checks.

0. Target Architecture

Samba AD DC:

lab2 100.100.100.102

LAB storage:

management SSH: 192.168.1.20:6953

NFS storage IP: 100.100.100.100

NFS server FQDN: lab-storage.lab.decs.internal

test export: /294t/dcloud/share/test_krb

LAB test client:

lab8 100.100.100.108

test mount: /mnt/lab-test-krb

identity provider: SSSD + adcli + AD RFC2307

Kerberos realm: LAB.DECS.INTERNAL

DNS domain: lab.decs.internal

NetBIOS domain: LAB

NFS server principal: nfs/lab-storage.lab.decs.internal@LAB.DECS.INTERNAL

Preferred NFS version: NFSv4.1

NFS security: sec=krb5

Default policy update as of 2026-07-06:

FARM and LAB should both default to `sec=krb5` for Kerberos NFS.
`sec=krb5` keeps Kerberos authentication while avoiding integrity/privacy wrapping overhead. Use `sec=krb5i` or `sec=krb5p` only as explicit opt-in modes for paths or tests that require those guarantees.

Historical note: the 2026-06-27 LAB8 PoC used `sec=krb5p` and passed. Keep those results as evidence, but do not treat `krb5p` as the new default.

Current status and latest evidence

The desired default for a new LAB Kerberos mount is `sec=krb5`. The LAB8 fault and regression investigations observed an existing operational/test mount using `NFSv4.1,sec=krb5p`; that is an observed test condition, not the canonical default. Verify the live export source and security flavor before changing any operational LAB mount.

Current isolated VM/kernel-matrix procedure: `../test/lab-kerberos-poc/vm-slot-regression/README.md` (tooling changes do not prove live configuration changes). Evidence, newest first:

- `../test/lab-kerberos-poc/results/2026-07-16-lab8-idmap-slot-repro/README.md`: reproduced an old storage-kernel NFSv4.1 idmap deferral/session-slot leak. Treat storage-kernel remediation as a rollout gate.
- `../test/lab-kerberos-poc/results/2026-07-10-lab8-nfs-gss-fault/README.md`: reproduced LAB5-like SSH D-state under NFS transport interruption and established bounded canary/forensics and retry-inhibition guidance.
- `../test/lab-kerberos-poc/results/lab8_test_krb_result_20260627.md`: records the original successful Samba AD/RFC2307/Kerberos write test.

The dated evidence contains two historical export spellings, `/294t/share/test-krb` and `/294t/dcloud/share/test_krb`. Do not infer the live source from the old filenames: check `exportfs -v` on storage and `findmnt` on the client, then use one path consistently.

If the already-created LAB test path is `/294t/share/test-krb` instead of `/294t/dcloud/share/test_krb`, use that path consistently in every storage export, mount, ownership, and verification command.

Important distinction from FARM:

FARM:

Synology NAS itself joins AD.

LAB:

The Linux storage server joins AD.

The `test_krb` directory/export does not "join" AD by itself.

1. Required Identity Invariant

For every LAB Kerberos container using the AD-backed `test_krb` path, the same Unix identity must be used everywhere:

```
DB uid/gid
= AD uidNumber/gidNumber
= storage server RFC2307 uid/gid view
= LAB client NFS returned uid/gid
= container uid/gid
```

Example:

```
username: test0524
group:    test0524_gid

DB:       10051:10051
AD user:   LAB\test0524 uidNumber=10051 gidNumber=10051
AD group:  LAB\test0524_gid gidNumber=10051
storage stat: 10051:10051
lab8 stat:  10051:10051
container: uid=10051 gid=10051
```

AD `primaryGroupID` matters. If a user remains in `Domain Users` as the primary group, new files may be created with the wrong group even when `gidNumber` is set. For DECS users, set the AD primary group to the intended private or shared DECS group.

2. Samba AD DC On lab2

LAB can reuse the FARM pattern, but with LAB values:

```
ad_realm: LAB.DECS.INTERNAL
ad_dns_domain: lab.decs.internal
ad_netbios_domain: LAB
ad_dc_fqdn: dc1.lab.decs.internal
ad_dc_service_ip: 100.100.100.102
storage_fqdn: lab-storage.lab.decs.internal
storage_ip: 100.100.100.100
```

The current LAB MIT KDC PoC may be removed for this test if it is not used by production. Do not run an independent MIT KDC and Samba AD DC for the same `LAB.DECS.INTERNAL` realm at the same time.

Required packages on lab2:

```
sudo apt-get update
sudo apt-get install -y acl attr dnsutils krb5-user samba winbind
```

Provision with RFC2307:

```
sudo systemctl disable --now smbd nmbd winbind 2>/dev/null || true

sudo samba-tool domain provision \
  --server-role=dc \
  --use-rfc2307 \
  --dns-backend=SAMBA_INTERNAL \
  --realm=LAB.DECS.INTERNAL \
  --domain=LAB \
  --host-name=dc1 \
  --host-ip=100.100.100.102

sudo systemctl enable --now samba-ad-dc
```

Install `/etc/krb5.conf` on lab2:

```
[libdefaults]
    default_realm = LAB.DECS.INTERNAL
    dns_lookup_realm = false
    dns_lookup_kdc = false
    rdns = false
    ticket_lifetime = 24h
    renew_lifetime = 7d
    forwardable = true

[realms]
    LAB.DECS.INTERNAL = {
        kdc = 100.100.100.102
        admin_server = 100.100.100.102
        default_domain = lab.decs.internal
    }

[domain_realm]
    .lab.decs.internal = LAB.DECS.INTERNAL
    lab.decs.internal = LAB.DECS.INTERNAL
```

Add DNS records in Samba internal DNS:

```
sudo samba-tool dns add 127.0.0.1 lab.decs.internal dc1 A 100.100.100.102 -U Administrator
sudo samba-tool dns add 127.0.0.1 lab.decs.internal lab-storage A 100.100.100.100 -U Administrator
```

Basic checks:

```
host -t SRV _kerberos._udp.lab.decs.internal 100.100.100.102
kinit Administrator@LAB.DECS.INTERNAL
klist
kdestroy
sudo samba-tool drs showrepl
```

3. AD User And Group Policy

For a user with no shared group:

```
AD user:          LAB\<username>
AD private group: LAB\<username>_gid
principal:        <username>@LAB.DECS.INTERNAL
uidNumber:        DB uid
gidNumber:        DB gid
primary group:    LAB\<username>_gid
```

For a user assigned to a shared group:

```
AD user:          LAB\<username>
AD private group: LAB\<username>_gid
AD shared group:  LAB\<groupname>
principal:        <username>@LAB.DECS.INTERNAL
uidNumber:        DB uid
gidNumber:        shared group gid
primary group:    LAB\<groupname>
```

Example:

```
sudo samba-tool group add test0524_gid || true
sudo samba-tool group addunixattrs test0524_gid 10051 || true

sudo samba-tool user add test0524 '<initial-password>' || true
sudo samba-tool user addunixattrs test0524 10051 10051 || true
sudo samba-tool group addmembers test0524_gid test0524 || true
sudo samba-tool user setprimarygroup test0524 test0524_gid
```

Verify:

```
sudo samba-tool user show test0524 | \
  grep -E '^(sAMAccountName|objectSid|uidNumber|gidNumber|primaryGroupID):'

sudo samba-tool group show test0524_gid | \
  grep -E '^(sAMAccountName|objectSid|gidNumber):'
```

4. LAB Storage Server Setup

The storage server is the AD domain member and the NFS server. Configure only the `test_krb` path first.

4.1 Packages

```
sudo apt-get update
sudo apt-get install -y \
  krb5-user nfs-kernel-server nfs-common \
  samba winbind libnss-winbind libpam-winbind
```

4.2 Kerberos Client Config

Install the LAB `/etc/krb5.conf` shown above. Ensure the storage host resolves:

```
getent hosts dc1.lab.decs.internal
getent hosts lab-storage.lab.decs.internal
```

4.3 Join Storage To AD

Recommended `/etc/samba/smb.conf` for the Linux storage domain member:

```
[global]
workgroup = LAB
realm = LAB.DECS.INTERNAL
security = ADS
kerberos method = secrets and keytab

winbind enum users = yes
winbind enum groups = yes
winbind use default domain = yes
winbind refresh tickets = yes
winbind nss info = rfc2307

idmap config * : backend = tdb
idmap config * : range = 300000-399999
idmap config LAB : backend = ad
idmap config LAB : schema_mode = rfc2307
idmap config LAB : range = 1000-299999
```

Required `/etc/nsswitch.conf` shape:

```
passwd:      files systemd winbind
group:       files systemd winbind
shadow:      files
gshadow:     files
```

Join:

```
sudo net ads join -U Administrator
sudo systemctl enable --now winbind
sudo systemctl restart winbind
```

Verify RFC2307 identity:

```
wbinfo -u | grep '^test0524$'
getent passwd test0524
getent group test0524_gid
id test0524
```

Expected numeric shape:

```
uid=10051(test0524) gid=10051(test0524_gid) groups=10051(test0524_gid)
```

4.4 NFS Server Principal And Keytab

Create the storage NFS service principal in AD and export a keytab usable by the storage NFS GSS service.

One workable pattern is to add the SPN to the storage machine account after the storage server joins AD:

```
sudo net ads keytab add nfs/lab-storage.lab.decs.internal -U Administrator
sudo klist -k /etc/krb5.keytab | grep nfs/lab-storage.lab.decs.internal
```

If `net ads keytab add` is not sufficient in the local Samba version, use `samba-tool spn add` on lab2 and export the keytab from AD, then install it as `/etc/krb5.keytab` on storage with mode `0600`.

Start NFS GSS services:

```
sudo systemctl enable --now nfs-server
sudo systemctl enable --now rpc-svcgssd 2>/dev/null || true
sudo systemctl restart nfs-server
sudo systemctl restart rpc-svcgssd 2>/dev/null || true
```

Depending on Ubuntu service names, `rpc-svcgssd` may be socket-activated or managed as part of `nfs-server`. Verify with:

```
ps -ef | grep -E 'svcgssd|rpc.svcgssd' | grep -v grep
```

5. test_krb Export

Create the test directory on storage:

```
sudo install -d -o root -g root -m 0755 /294t/dcloud/share/test_krb
sudo install -d -o root -g root -m 0755 /294t/dcloud/share/test_krb/user-share
```

Add an export for LAB test clients only:

```
/294t/dcloud/share/test_krb \
100.100.100.108(rw,async,no_subtree_check,crossmnt,root_squash,sec=krb5)
```


Apply:

```
sudo exportfs -ra
sudo exportfs -v | grep -A2 '/294t/dcloud/share/test_krb'
```

Flush NFS/RPC caches after identity or export changes:

```
now=$(date +%s)
for p in \
  /proc/net/rpc/auth.unix.gid/flush \
  /proc/net/rpc/nfs4.idtoname/flush \
  /proc/net/rpc/nfs4.nametoid/flush \
  /proc/net/rpc/auth.rpcsec.init/flush \
  /proc/net/rpc/auth.rpcsec.context/flush
do
  [ -w "$p" ] && printf "%s\n" "$now" | sudo tee "$p" >/dev/null || true
done
```

6. LAB Client Setup

Every LAB host that mounts the AD-backed test export needs Kerberos, NFS, and AD/RFC2307 identity lookup. Ordinary LAB clients use SSSD with an `adcli` machine join. Do not copy another server's `/etc/sss/sss.conf`; apply the role-based configuration below and verify the local machine join.

Samba AD DCs and the LAB storage server have separate winbind-based role configuration described earlier in this runbook. The SSSD procedure in this section is for ordinary, non-DC LAB clients.

Packages:

```
sudo apt-get update
sudo apt-get install -y \
  keyutils krb5-user nfs-common \
  adcli libnss-sss libpam-sss realmd sssd-ad sssd-tools
```

Install the LAB `/etc/krb5.conf` shown in section 3, then check whether the host's LAB machine join is still valid:

```
sudo adcli testjoin \
  --domain=lab.decs.internal \
  --domain-controller=dc1.lab.decs.internal
```

If `adcli testjoin` fails, join the local host to LAB AD. Read the password without placing it in shell history:

```
read -rsp 'LAB AD Administrator password: ' DECS_LAB_AD_PASSWORD
printf '\n'
printf '%s\n' "$DECS_LAB_AD_PASSWORD" | sudo adcli join lab.decs.internal \
  --domain-realm=LAB.DECS.INTERNAL \
  --domain-controller=dc1.lab.decs.internal \
  --login-user=Administrator \
  --stdin-password \
  --host-fqdn="$(hostname -s | tr '[:upper:]' '[:lower:]').lab.decs.internal" \
  --computer-name="$(hostname -s | tr '[:lower:]' '[:upper:]')"
unset DECS_LAB_AD_PASSWORD
sudo adcli testjoin \
  --domain=lab.decs.internal \
  --domain-controller=dc1.lab.decs.internal
```

Configure `/etc/sss/sss.conf` with mode `0600`:

```
[sss]
config_file_version = 2
services = nss, pam, ssh
domains = lab.decs.internal

[domain/lab.decs.internal]
ad_domain = lab.decs.internal
krb5_realm = LAB.DECS.INTERNAL
id_provider = ad
access_provider = permit
ad_server = dc1.lab.decs.internal, lab3.lab.decs.internal, lab4.lab.decs.internal
krb5_server = dc1.lab.decs.internal, lab3.lab.decs.internal, lab4.lab.decs.internal
ldap_id_mapping = False
use_fully_qualified_names = False
fallback_homedir = /home/%u
default_shell = /bin/bash
cache_credentials = True
krb5_store_password_if_offline = True
ad_gpo_access_control = permissive
dyndns_update = False
```

`ldap_id_mapping = False` is required: the UID/GID must come from the AD RFC2307 `uidNumber` / `gidNumber` attributes instead of being generated from the AD SID.

Back up NSS configuration before enabling SSSD. An ordinary LAB client should resolve users and groups through `sss`, not through the standalone winbind configuration that may remain after a server role change:

```
sudo cp -a /etc/nsswitch.conf \
"/etc/nsswitch.conf.decs-pre-sssd-${date +%Y%m%d%H%M%S}"
sudo sed -i -E \
-e 's/^(passwd:).*/\1      files systemd sss/' \
-e 's/^(group:).*/\1      files systemd sss/' \
-e 's/^(shadow:).*/\1     files sss/' \
/etc/nsswitch.conf
sudo chmod 0600 /etc/sss/sss.conf
sudo sssctl config-check
sudo systemctl disable --now winbind 2>/dev/null || true
sudo systemctl enable --now sssd
sudo sss_cache -E
sudo systemctl restart sssd
```

Expected NSS lines on an ordinary LAB client:

```
passwd:      files systemd sss
group:       files systemd sss
shadow:      files sss
```

Verify RFC2307 identity lookup before mounting or provisioning containers:

```
systemctl is-active sssd
getent passwd test0524
getent group test0524_gid
id test0524
```

The numeric UID/GID returned here must match AD `uidNumber` / `gidNumber` and the DB values. A successful Kerberos machine join by itself does not prove NSS identity resolution is correct.

Start `rpc.gssd`:

```
sudo systemctl enable --now rpc-gssd
sudo systemctl restart rpc-gssd
```

Mount the test export:

```
sudo install -d -o root -g root -m 0755 /mnt/lab-test-krb
sudo mount -t nfs4 \
-o vers=4.1,sec=krb5,hard,proto=tcp,_netdev \
lab-storage.lab.decs.internal:/294t/dcloud/share/test_krb \
/mnt/lab-test-krb
```

Verify mount options:

```
findmnt /mnt/lab-test-krb -o TARGET,SOURCE,FSTYPE,OPTIONS
```

7. Container Provisioning Model

For LAB AD mode, `create-container --enable-kerberos` should eventually do the same logical work as FARM:

1. Reserve or reuse DB uid/gid.
2. Ensure AD private/shared group exists with gidNumber.
3. Ensure AD user exists with uidNumber/gidNumber.
4. Set AD primary group to the intended DECS group.
5. Ensure storage home exists under test_krb/user-share/<username>.
6. Chown storage home to numeric uid/gid.
7. Create or rotate user keytab as a root-only secret.
8. Create host ccache at /run/user/<uid>/krb5cc.
9. Bind mount ccache into the container.
10. Set KRB5CCNAME=FILE:/run/user/<uid>/krb5cc.

The container must not be the authority for identity. The authority is:

```
DB + AD RFC2307 + storage winbind/NSS + client SSSD/NSS
```

8. Verification Checklist

8.1 AD And Identity

On lab2:

```
sudo samba-tool user show test0524 | grep -E 'uidNumber|gidNumber|primaryGroupID'
sudo samba-tool group show test0524_gid | grep gidNumber
```

On storage and the LAB client (winbind/NSS on storage, SSSD/NSS on the client):

```
getent passwd test0524
getent group test0524_gid
id test0524
```

8.2 NFS Kerberos

On storage:

```
sudo exportfs -v | grep -A2 test_krb
sudo klist -k /etc/krb5.keytab | grep nfs/lab-storage.lab.decs.internal
```

On lab8:

```
findmnt /mnt/lab-test-krb -o TARGET,SOURCE,FSTYPE,OPTIONS
```

The mount options must contain:

```
vers=4.1
sec=krb5
```

8.3 User Write

With a valid test user ticket/ccache:

```
sudo -u test0524 KRB5CCNAME=FILE:/run/user/10051/krb5cc \
  bash -lc 'touch /mnt/lab-test-krb/user-share/test0524/write_probe && stat -c "%u:%g %U:%G %A %n" /mnt/lab-
test-krb/user-share/test0524/write_probe'
```

Expected numeric result:

```
10051:10051 test0524:test0524_gid
```

8.4 Group Sharing

Create two users in the same AD group, set that group as primary for both, and verify that files created through NFS land in the shared group. Then verify user-managed sharing inside home:

```
mkdir -p ~/sharing_dir
chmod 2770 ~/sharing_dir
```

Another user in the same AD primary/shared group should be able to read/write when Unix mode bits permit it. A user outside that AD group should fail.

8.5 Container Services

For a test container using the AD-backed test mount:

```
SSH login succeeds.  
Jupyter HTTP returns 200.  
noVNC HTTP returns 200.  
KRB5CCNAME points to /run/user/<uid>/krb5cc.  
User can write its own home.  
Cross-home write fails without group permission.  
Restricted sudo prevents UID spoofing/root shell paths.
```

9. Rollback

For a test-only rollout, rollback should be simple:

```
# lab8  
sudo umount /mnt/lab-test-krb  
  
# storage  
sudo exportfs -u 100.100.100.108:/294t/dcloud/share/test_krb  
sudo exportfs -ra
```

If the LAB Samba AD DC was only for PoC and no production dependency exists:

```
sudo systemctl disable --now samba-ad-dc
```

Do not delete AD database files or keytabs until the test result and rollback scope are reviewed.

10. Notes Compared With FARM

LAB does not need:

```
synowin -joinDomain  
Synology DSM UI changes  
pkg-synosamba-winbindd.service  
Synology-specific SMBService net cache flush paths
```

LAB does need:

```
Linux storage AD member join
storage-side winbind RFC2307 mapping
storage-side NFS server principal/keytab
test_krb export with sec=krb5 by default
ordinary LAB clients with SSSD/adcli RFC2307 NSS and rpc.gssd
```

The biggest consistency benefit is that FARM and LAB can eventually share the same DECS rule:

```
DB uid/gid = AD RFC2307 uid/gid = storage/NFS uid/gid = container uid/gid
```

The biggest operational risk is replacing the existing LAB MIT Kerberos PoC with Samba AD for the same realm. This is acceptable if the MIT PoC is not production, but the replacement should be done deliberately and verified on `test_krb` before touching wider LAB mounts.

LAB keytab 상태도 FARM과 같은 읽기 전용 checker로 확인한다. 별도 LAB 스크립트는 없으며 endpoint와 principal만 `config/lab.env`에 정의한다.

```
/home/jy/server_manage/monitoring/health-checks/kerberos-nfs-keytab/script/check-nfs-keytab.sh --profile lab
```

LAB은 아직 PoC이므로 이 명령은 필요할 때 직접 실행하고, `check-nfs-keytab@lab.timer`는 LAB이 상시 운영 대상이 된 뒤 활성화한다.

Kerberos Ticket Refresh Margin

Updated on 2026-06-29.

DECS containers do not receive user keytabs. Each LAB Docker host keeps the user keytab as a root-only secret under `/etc/decs-krb/keytabs` and exposes only the host ccache at `/run/user/<uid>/krb5cc` to the container.

Ticket refresh is handled by:

```
/usr/local/sbin/decs-krb-refresh
/etc/systemd/system/decs-krb-refresh@.service
/etc/systemd/system/decs-krb-refresh@.timer
/etc/decs-krb/refresh.d/<username>.env
```

The timer normally runs every 1 hour. The refresh script first inspects the existing ccache. If the Kerberos `renew until` deadline is more than 24 hours away, it performs `kinit -R`. If `renew until` is within 24 hours, missing, or the renewal fails, it uses the root-only keytab to issue a fresh ticket:

```
kinit -k -t /etc/decs-krb/keytabs/<username>.keytab \  
-c FILE:/run/user/<uid>/krb5cc \  
<username>@LAB.DECS.INTERNAL
```

The 24-hour margin is controlled by `DECS_KRB_REISSUE_BEFORE_SECONDS` and defaults to `86400`. It can be overridden per user by adding the variable to `/etc/decs-krb/refresh.d/<username>.env`.

Operational checks:

```
systemctl list-timers 'decs-krb-refresh@*.timer'  
systemctl status decs-krb-refresh@<username>.timer  
systemctl start decs-krb-refresh@<username>.service  
klist -c FILE:/run/user/<uid>/krb5cc
```